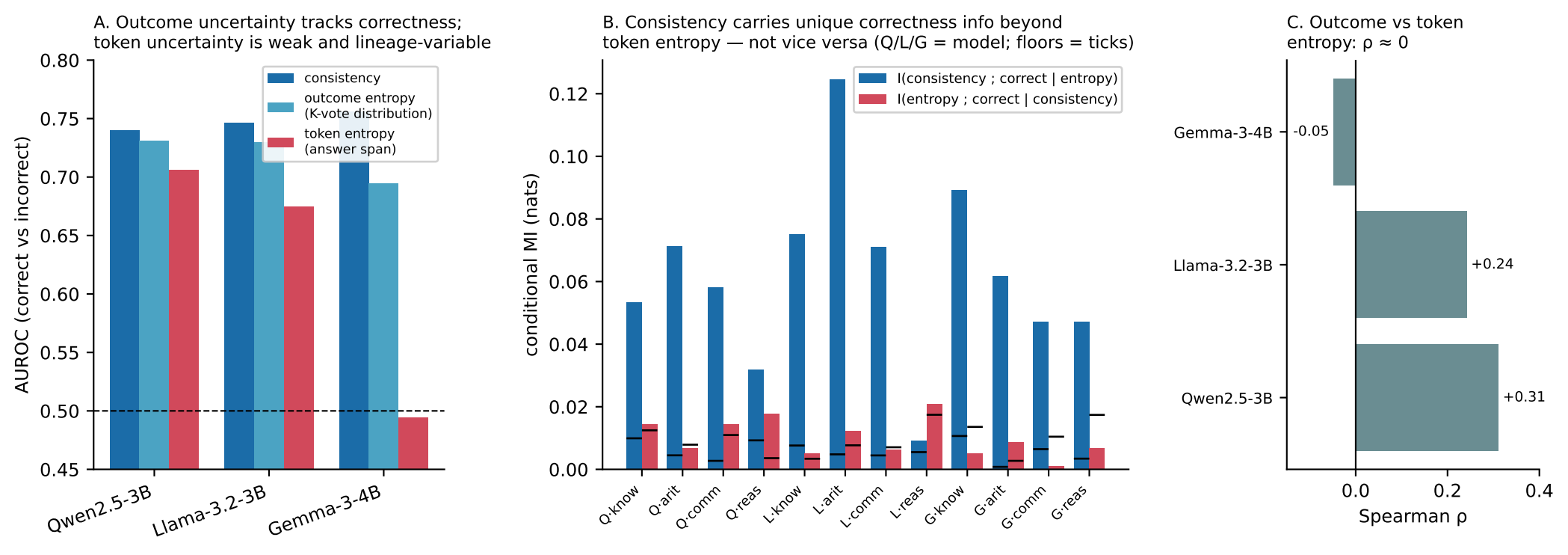
A. Outcome uncertainty tracks correctness;
token uncertainty is weak and lineage-variable
AUROC (correct vs incorrect)
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
consistency
outcome entropy
(K-vote distribution)
token entropy
(answer span)
Qwen2.5-3B
Llama-3.2-3B
Gemma-3-4B
B. Consistency carries unique correctness info beyond
token entropy — not vice versa (Q/L/G = model; floors = ticks)
conditional MI (nats)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
I(consistency ; correct | entropy)
I(entropy ; correct | consistency)
Q·know
Q·arit
Q·comm
Q·reas
L·know
L·arit
L·comm
L·reas
G·know
G·arit
G·comm
G·reas
C. Outcome vs token
entropy: ρ ≈ 0
Gemma-3-4B
Llama-3.2-3B
Qwen2.5-3B
-0.05
+0.24
+0.31
0.0
0.2
0.4
Spearman ρ